STUDENT PROJECT PROGRESS TRACKER

A PROJECT REPORT

Submitted by

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in partial fulfilment of the End Semester Examination for the course

CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY

Artificial Intelligence and Machine Learning



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

2026–2027

RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this project report titled “STUDENT PROJECT PROGRESS TRACKER” is the bonafide work of THAVANESHWARAN S, (2116241501231), who carried out the individual project work on the Zoho Creator low-code platform for the course **CB23F35 – Low-Code Application Development using Zoho Creator**, during the academic year 2026–2027.

SIGNATURE	**SIGNATURE**
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Submitted for the End Semester Examination (Individual Project) held on [DATE].

INTERNAL EXAMINER	**EXTERNAL EXAMINER**
Signature: _____	Signature: _____

DECLARATION

I, THAVANESHWARAN S (2116241501231), hereby declare that the project report titled “STUDENT PROJECT PROGRESS TRACKER”, submitted for the End Semester Examination of CB23F35 – Low-Code Application Development using Zoho Creator, is a record of the project work carried out by me. The report is intended to describe the design and development of a student project management application using Zoho Creator. Details that require confirmation from the live application, including final screenshots, exact field configurations, workflows, permissions and deployment status, will be verified before submission.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Any external references and AI-assisted development support used in preparing the application or report will be acknowledged appropriately.

Place: Chennai

Date: _____

SIGNATURE OF THE STUDENT

THAVANESHWARAN S– 2116241501231

ACKNOWLEDGEMENT

I express my sincere gratitude to the management of Rajalakshmi Engineering College for providing the infrastructure and learning environment required to undertake this project.

I thank the Principal of Rajalakshmi Engineering College for encouraging students to develop practical, industry-relevant applications as part of their academic learning.

I express my gratitude to the Head of the Department of Artificial Intelligence and Machine Learning for the guidance and support provided throughout the course.

I sincerely thank my course faculty for valuable guidance, feedback and suggestions during the planning and development of this application and report.

I also acknowledge Zoho Creator and its learning resources for providing a low-code platform through which forms, reports, pages and workflows can be configured. Finally, I thank my family and friends for their encouragement and support.

THAVANESHWARAN S

ABSTRACT

Student teams often manage academic projects through a combination of spreadsheets, documents, messaging applications and verbal discussions. Project details may be stored separately from task assignments, deadlines, milestone records and mentor feedback. This fragmented approach can make it difficult to identify incomplete work, track responsibilities, review progress consistently and recognise delays before submission dates. The Student Project Progress Tracker is proposed as a centralised application for organising academic project information and supporting the project lifecycle from initial planning to final completion.

The application is planned for development on Zoho Creator using its form builder, reports, pages and workflow capabilities, with Zia AI used as an AI-assisted development aid where supported. The proposed data-entry modules include Student Profile, Project, Task, Milestone, Progress Update and Meeting Notes. These modules are intended to connect related records through lookup fields so that tasks, milestones, progress entries and meeting decisions can be associated with the relevant project and students. Seven planned pages provide dashboard summaries, project management, task management, project details, progress tracking, team workspace and reports and analytics. The design aims to help students understand their responsibilities and help project guides review progress from a single workspace.

The report documents the problem definition, requirements, system design, data model, planned implementation, testing approach, security considerations and future extensions. Because final configuration and live application evidence must be confirmed from the actual Zoho Creator application, screenshot areas and configuration-specific details are intentionally left as placeholders. Workflow automation, calculations, notifications, role-based access and analytics should be described as operational only after they have been configured and tested. The proposed system is expected to improve the organisation of project information and reduce dependence on manual status compilation, while its effectiveness will depend on accurate record entry and verified configuration.

Keywords: Low-Code Development; Zoho Creator; Student Project Management; Task Tracking; Workflow Automation; Progress Analytics.

PROJECT AT A GLANCE

Attribute	Details
Project Title	Student Project Progress Tracker
Application Domain	Academic project planning and team coordination
Student Name & Register No.	THAVANESHWARAN S – 2116241501231
Development Platform	Zoho Creator
AI-assisted Development	Zia AI, where supported and used
Application Name / Link Name	[Enter exact application name and link name after verification]
Published / Live URL	[Insert only if published]
Demo Credentials	[Do not include passwords in the report; provide demo accounts securely if required]
Forms Planned	6: Student Profile, Project, Task, Milestone, Progress Update, Meeting Notes
Pages Planned	7 dashboard and management pages
Reports / Charts	Project status, tasks, deadlines, progress history and milestone reports; exact count to be confirmed
Workflows	Validation, progress calculation, overdue identification and notifications are planned; confirm those implemented
Deluge Functions	[List only verified scripts after implementation]
Integrations	No external integration confirmed in the supplied project description
Roles	Student, Project Guide, Administrator — permissions to be verified
Environment / Deployment	[Record Development / Stage / Production status after verification]
Deployment Date	[Enter if deployed]

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
FR	Functional Requirement
NFR	Non-Functional Requirement
LCAP	Low-Code Application Platform
RBAC	Role-Based Access Control
UI	User Interface
UAT	User Acceptance Testing
Zoho Creator	Low-code application development platform

CHAPTER 1

INTRODUCTION

1.1 Background

Academic projects are an important part of engineering education. They require students to define a problem, plan activities, divide work among team members, develop and test a solution, record milestones and submit a final deliverable within a specified period. A project guide may need to review several teams at different stages and provide timely feedback. When project information is scattered across documents, spreadsheets, chats and individual notes, it becomes difficult to maintain a reliable view of the complete project.

The Student Project Progress Tracker is designed to provide a common workspace for recording project information and following progress. The application is intended for college students, project teams, faculty mentors and administrators. It uses Zoho Creator as the low-code platform so that structured forms, reports, pages and workflow logic can be configured without building an entire web application from scratch.

1.2 Problem Statement

The problem addressed by this project is the lack of a single, structured place to manage academic project details, task ownership, milestones, progress updates and meeting decisions. A fragmented process makes it harder for students and guides to determine the current status of work, identify blockers and respond to approaching deadlines.

- Project information and team details may be duplicated across separate files.
- Task ownership, priorities and due dates may not be visible to all relevant team members.
- Progress updates and mentor feedback may be difficult to retrieve chronologically.
- Delayed tasks and milestones may be identified only during periodic manual reviews.
- Preparing a consolidated project status report may require repeated manual work.

1.3 Objectives

- Design structured forms to capture student profiles, projects, tasks, milestones, progress updates and meeting notes.
- Relate project records using lookup fields so that associated tasks and updates can be retrieved consistently.
- Provide project and task views that make ownership, status, priorities and deadlines easy to review.
- Organise progress history, milestone information and meeting decisions for student teams and guides.
- Design reports and dashboard summaries for project status, overdue work and upcoming deadlines.
- Evaluate validation, access permissions and workflow behaviour through documented test cases before claiming production readiness.

1.4 Scope of the Project

The planned scope includes the six data-entry forms, related project and student lookups, seven application pages, reports for project and task monitoring, and optional workflow logic for validation, progress calculation, overdue identification and deadline notifications. The primary access channel is the Zoho Creator web application. The intended users are students, project guides and administrators.

The initial scope does not assume external integrations, a mobile-specific implementation, advanced AI risk prediction, GitHub activity synchronisation or a fully deployed production environment. Such features should be included only if they are later implemented and verified. Access-control rules, notification delivery and any AI-generated outputs require independent testing.

1.5 Justification for a Low-Code Approach

Zoho Creator is suitable for a prototype of this type because its visual builders support data-entry forms, lookup relationships, reports, dashboards and workflow configuration within one environment. A low-code approach can reduce the amount of custom infrastructure code and make routine form or report changes easier to maintain. Deluge can

be introduced for business rules and event-driven automation when built-in configuration is insufficient. These benefits do not remove the need for sound data modelling, validation, security configuration and testing. Platform features described in this report are not automatically evidence that the corresponding feature has been enabled in this particular application [1]–[3].

1.6 Organisation of the Report

Chapter 2 studies the existing workflow and low-code development approach. Chapter 3 defines stakeholders and requirements. Chapter 4 presents the proposed architecture, data model, workflows and access design. Chapter 5 describes the application components and identifies screenshot locations. Chapter 6 provides a test strategy and test cases. Chapter 7 discusses deployment, security and governance. Chapter 8 evaluates the intended outcomes and limitations. Chapter 9 concludes the report and lists future enhancements.

CHAPTER 2

LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

The baseline considered for this project is a typical student-team workflow in which project details are stored in a document or spreadsheet, task assignments are shared through messaging applications, progress is reported during meetings, and the guide collects updates when needed. This baseline is used as a design assumption; it is not a measured audit of a specific department.

Project idea and scope → team members and guide identified → tasks and deadlines shared → team members work independently → progress discussed in meetings → guide requests updates → status report compiled manually → final submission.

2.2 Limitations of the Existing System

- Information can be duplicated or become inconsistent when multiple copies of project records exist.
- Team members may have incomplete visibility of the tasks assigned to others.
- Meeting decisions and blockers may be lost in message histories.
- There may be no automatic view of overdue tasks or upcoming milestones.
- Progress reports depend on timely manual updates and compilation.

2.3 Low-Code versus Traditional Development

Criterion	**Traditional Development**	**Zoho Creator Low-Code Approach**
Application structure	Separate frontend, backend, database and hosting components may need to be built.	Forms, reports, pages and data storage are configured within the platform.
Development effort	Requires implementation and maintenance of more infrastructure code.	Visual configuration reduces boilerplate; data design and logic still require care.
Business rules	Implemented in application code and backend services.	Can use field rules, workflows and Deluge where supported.
Reporting	Requires custom reports or a separate analytics tool.	Built-in reports and charts can present records.
Customisation	Broad architectural and UI control.	Customisation is bounded by Creator capabilities and plan limits.

Security	Must implement and maintain authentication and authorisation components.	Platform controls are available but must be configured and tested.
Cost and effort	Depends on hosting, engineering time and maintenance.	Depends on subscription, configuration and any additional services; no cost comparison was measured.

2.4 Literature Survey

The following sources provide background for the selected development approach. The report distinguishes general platform documentation from claims about features actually configured in the project.

S.No.	Source	Topic	Key point	Relevance
1	Zoho Corporation [1]	Zoho Creator Help	Documents platform configuration for forms, reports and applications.	Reference for implementation concepts.
2	Zoho Corporation [2]	Deluge Reference	Describes scripting constructs and supported tasks.	Reference for optional custom business logic.
3	Zoho Corporation [3]	Creator Resources	Provides learning material and implementation examples.	Supports learning and troubleshooting.
4	Low-code development literature [4]	Low-code platforms	Low-code tools can accelerate application delivery but introduce platform constraints.	Supports qualitative comparison.
5	Software engineering practice [5]	Requirements and testing	Requirements should be traceable to tests and evidence.	Supports testable FRs and validation.

2.5 Emerging Trends in Low-Code Development

Relevant trends include AI-assisted application creation, citizen development, workflow automation and composable applications. AI-assisted tools can help scaffold forms or suggest configurations, but generated components must be reviewed for correctness. Workflow automation can reduce repeated manual actions when triggers and conditions are explicitly defined. The project follows a staged approach: establish the data structure first, connect reports and pages second, and introduce automation only after the underlying records are reliable.

2.6 Proposed System

The proposed system centralises project information in Zoho Creator. Student profiles supply team information; the Project form stores the primary project record; Task and Milestone forms break work into trackable units; Progress Update records capture completed work, next steps and blockers; and Meeting Notes preserve decisions and action items. Pages and reports are intended to provide a concise view of project status and upcoming work. The actual implementation status of each component must be verified against the live application before final submission.

CHAPTER 3

REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Role	Description	Responsibilities
Student	Member of one or more academic project teams.	View authorised projects; update assigned tasks; submit progress updates; review meeting actions.
Project Guide	Faculty mentor supervising projects.	Review project progress, milestones, blockers and submitted updates; provide feedback.
Administrator	Application maintainer.	Manage configuration and student records; maintain forms, reports and permissions.

3.2 Functional Requirements

ID	Requirement	Primary role	Priority
FR-01	The system shall capture student profile details and skills.	Student/Admin	High
FR-02	The system shall create a project record with title, description, dates, team, guide and technology details where fields are configured.	Student/Admin	High
FR-03	The system shall record tasks with project association, assignee, priority, status and due date.	Student	High
FR-04	The system shall record milestones associated with a project.	Student/Guide	High
FR-05	The system shall allow students to submit progress updates describing completed work, next steps and blockers.	Student	High
FR-06	The system shall store meeting notes, decisions and action items.	Student/Guide	Medium
FR-07	The system shall provide reports for projects, tasks, deadlines and progress history when configured.	Student/Guide/Admin	High
FR-08	The system shall identify overdue tasks based on due date and non-completed	Guide/Admin	Medium

	status if the corresponding logic is implemented.		
FR-09	The system shall validate mandatory fields and date relationships.	All permitted users	High
FR-10	The system shall restrict data access according to configured permissions.	All roles	High

3.3 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Usability	Forms and pages should use consistent labels and a clear navigation structure.
NFR-02	Data integrity	Related records should use lookup fields rather than relying solely on manually typed names.
NFR-03	Performance	Reports should use appropriate filters and avoid unnecessary columns as record volumes grow.
NFR-04	Security	Users should access only records and actions authorised by configured Creator permissions.
NFR-05	Maintainability	Forms, reports, fields and workflows should use consistent names and documented purposes.
NFR-06	Reliability	Workflow and notification behaviour should be tested for valid, invalid and boundary inputs.
NFR-07	Compatibility	The web interface should be tested in a supported modern browser.

3.4 Hardware and Software Requirements

Category	Specification
Development platform	Zoho Creator account with access to application builder features.
Client device	Desktop or laptop with a modern web browser.
Browser	Current Google Chrome, Microsoft Edge or another supported browser.
Connectivity	Internet connection for accessing the hosted Creator application.
Scripting	Deluge only where custom logic is required and implemented.
Documentation	Microsoft Word or compatible editor for the report.

3.5 Use Case Model

The principal actors are Student, Project Guide and Administrator. The student records project work and progress. The guide reviews progress and milestones. The administrator maintains application configuration and access permissions.

Use Case ID	**UC-01**
Use Case Name	Submit Progress Update
Actor(s)	Student
Pre-condition	The student has authorised access and the relevant project record exists.
Main Flow	1. Open Progress Tracking. 2. Select the associated project. 3. Enter completed work, next steps and blockers. 4. Submit the form. 5. Verify the saved record in the relevant report.
Alternate Flow	If mandatory fields are missing, the form should show validation feedback and should not save an incomplete record.
Post-condition	A progress update is stored and linked to the selected project, provided the form and lookup are configured.

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

The proposed architecture has four logical layers. The presentation layer contains the Zoho Creator pages and reports accessed by students, guides and administrators. The application layer contains forms, lookup relationships, validation rules and configured workflows. The data layer stores student, project, task, milestone, progress and meeting records. Optional notification or external integration services should be included only if they are configured and verified.

4.2 Data Model

The Project record acts as the central entity. Task, Milestone, Progress Update and Meeting Notes records should reference their parent project using lookup fields. Student Profile records provide information about students and can be referenced for team membership or task assignment. Exact cardinalities depend on the fields configured in the live app.

Entity / Form	Key information	Related entity	Relationship	Implementation approach
Student_Profile_Form	Student identity, department, email, skills	Projects, tasks, meetings	One student may relate to multiple records	Lookup fields where required
Project_Form	Title, description, category, dates, guide, team	Tasks, milestones, progress, meetings	One project to many child records	Project lookup in child forms
Task_Form	Task title, assignee, priority, status, due date, blockers	Project and student	Many tasks per project	Project and student lookups
Milestone_Form	Milestone name, target date, status	Project	Many milestones per project	Project lookup
Progress_Update_Form	Update date, completed work, next steps, blockers	Project and student	Many updates per project/student	Project and student lookups
Meeting_Notes_Form	Meeting date, attendees, decisions, action items	Project and student attendees	Many meetings per project	Project lookup and attendee fields

4.3 Form Design

The application is planned around six forms. The exact field link names, field types, mandatory settings, validation rules and defaults must be transcribed from the live Creator Form Builder after the final version is available. The following tables specify the logical fields expected for the design and should be reconciled with actual implementation.

Form	Example fields	Suggested field types	Validation / notes
Student Profile	Name, email, department, skills	Single Line, Email, Dropdown, Multi Line	Require essential identity fields; validate email format.
Project	Title, description, category, start date, deadline, guide, team, technology stack, repository URL, status	Single Line, Multi Line, Dropdown, Date, Lookup, URL	Deadline should not precede start date.
Task	Task title, project, assignee, priority, status, due date, completion %, blocker description	Single Line, Lookup, Dropdown, Date, Percent/Number, Multi Line	Completion value should be within the configured range.
Milestone	Milestone title, project, target date, status, completion date	Single Line, Lookup, Date, Dropdown	Status and dates should be consistent.
Progress Update	Project, student, update date, completed work, next steps, challenges, help needed	Lookup, Date, Multi Line, Dropdown	Require project and update summary.
Meeting Notes	Project, date, attendees, discussion, decisions, action items, next meeting	Lookup, Date, Multi Line	Capture actionable decisions and owners.

4.4 Process and Workflow Design

The core process begins when a project is created. Tasks and milestones are then associated with that project. Team members update task status and submit progress updates. The guide reviews the latest information and identifies blockers. If workflow automation is configured, it may calculate progress, flag overdue tasks or send deadline notifications. These automated behaviours must be verified in Creator before being reported as implemented.

4.5 User Interface and Navigation Design

The planned navigation consists of Student Dashboard, Project Management, Task Management, Project Details, Progress Tracking, Team Workspace, and Reports and Analytics. The interface is intended to use a consistent blue accent theme and concise

summaries so users can identify active projects, incomplete tasks, overdue work and approaching deadlines without opening multiple records.

4.6 Access Control Design

The following matrix is a proposed access design, not evidence that permissions have already been configured. Zoho Creator user, profile and record-level permissions must be checked in the live application. Hiding a page or button alone is not a substitute for data-level access control.

Component	Administrator	Project Guide	Student
Student profiles	A/E/V/D	V for assigned projects as permitted	V own profile; E own profile if allowed
Projects	A/E/V/D	V/E for assigned projects	V assigned projects; create/edit only if allowed
Tasks	A/E/V/D	V/E for assigned projects	V/E assigned tasks
Milestones	A/E/V/D	V/E for assigned projects	V; update permitted fields
Progress updates	A/E/V/D	V; feedback if implemented	A/V own updates
Meeting notes	A/E/V/D	A/E/V for assigned projects	A/V for team projects as permitted
Reports and dashboards	V all authorised data	V assigned projects	V own/team data as configured

CHAPTER 5

IMPLEMENTATION

This chapter documents the Student Project Progress Tracker as shown in the supplied Zoho Creator screenshots. The figures below provide visual evidence of the dashboard, data-entry forms, reports and status boards visible in the application. Screenshots not supplied for other modules remain to be added after those screens are available.

5.1 Application Setup

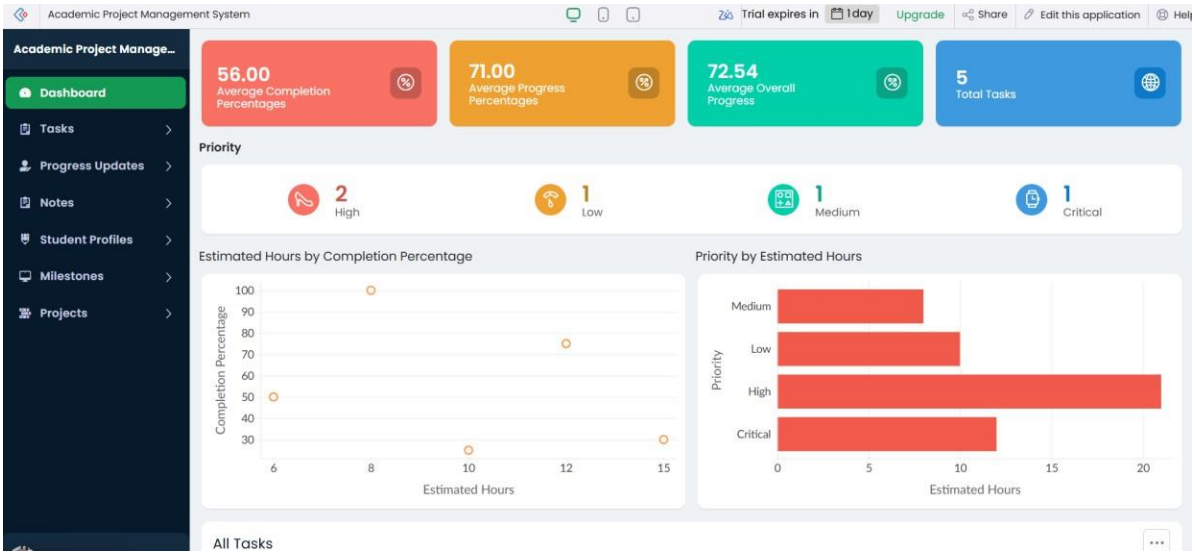


Figure 5.2: Academic Project Management System dashboard with project summary metrics.

The application is intended to be maintained in Zoho Creator. The setup process includes creating the application, arranging navigation sections, establishing consistent form and report names, and applying the selected visual theme. The exact application link name, Creator edition, publication status and navigation labels should be recorded from the live application rather than inferred.

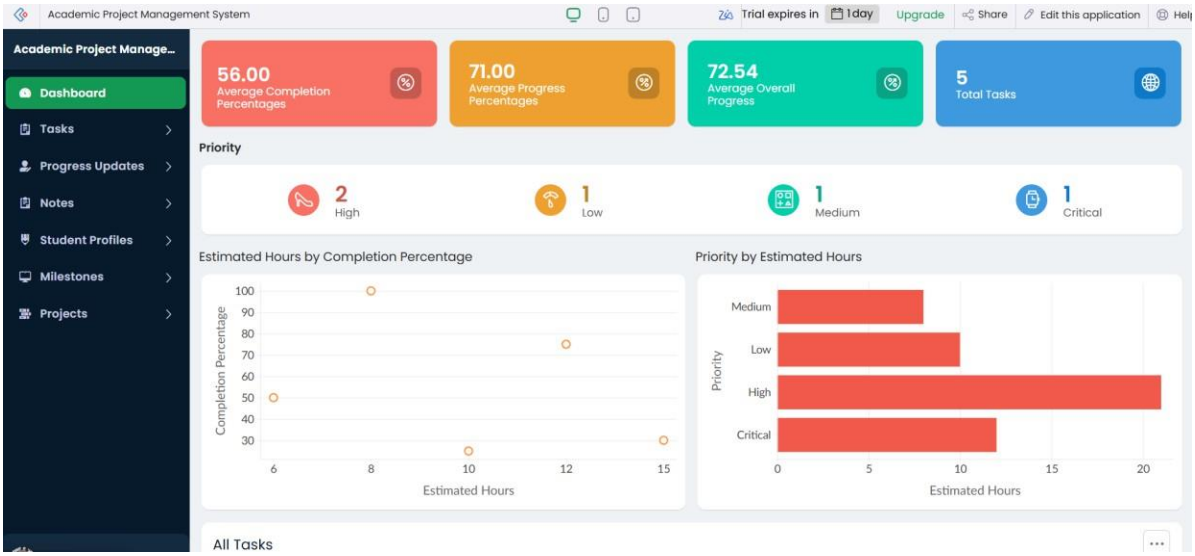


Figure 5.1: Academic Project Management System dashboard.

5.2 Forms and Data Capture

Student_Profile_Form

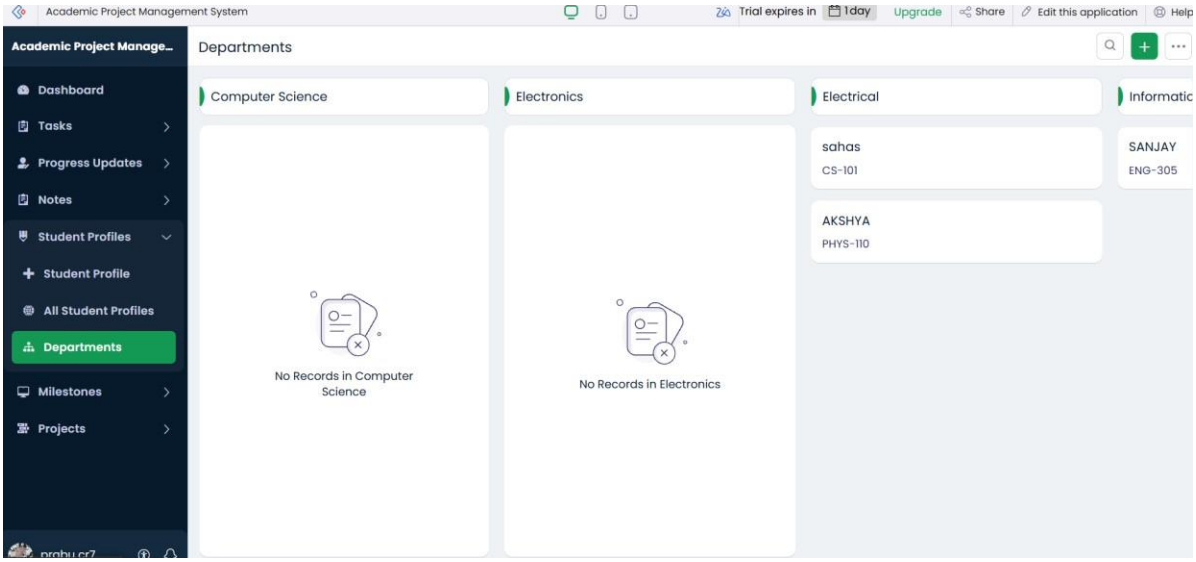


Figure 5.8: Departments board grouping student records by department.

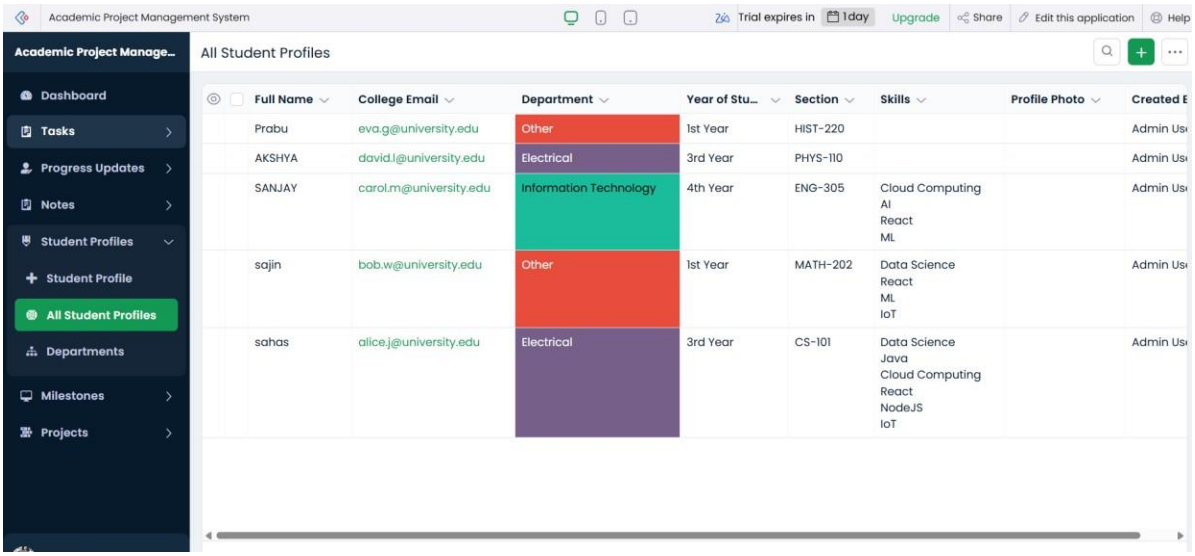


Figure 5.7: All Student Profiles report with student and department information.

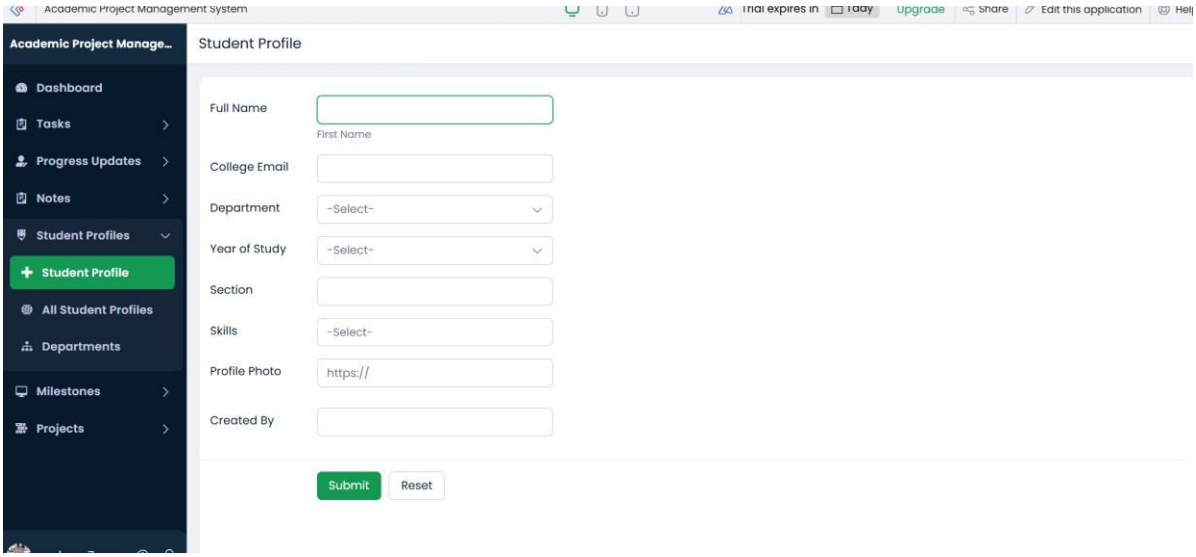


Figure 5.6: Student Profile form for capturing student details.

Captures student details and skills that can be referenced by project and task records.

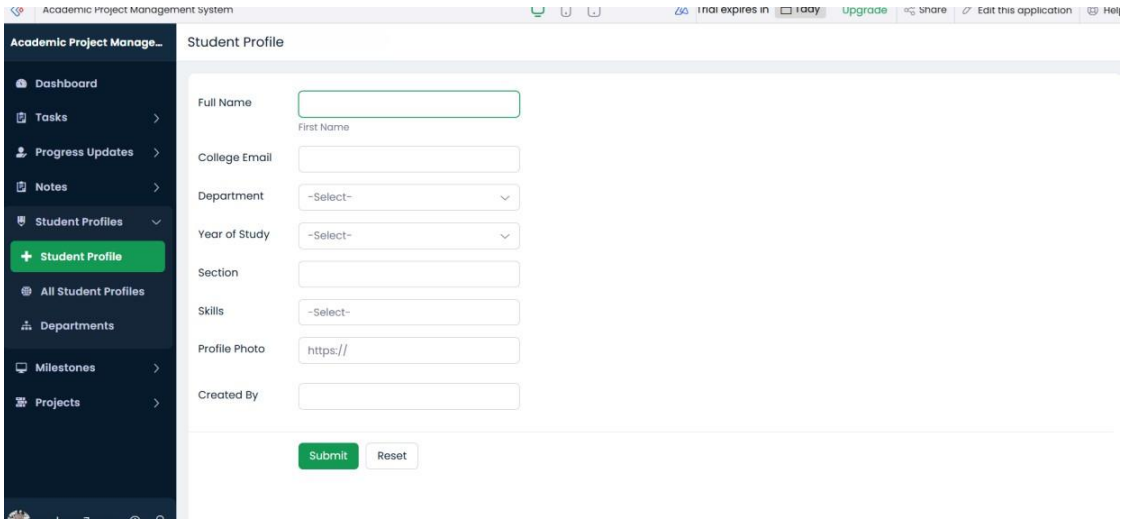


Figure 5.2: Student Profile form in Zoho Creator.

Project_Form

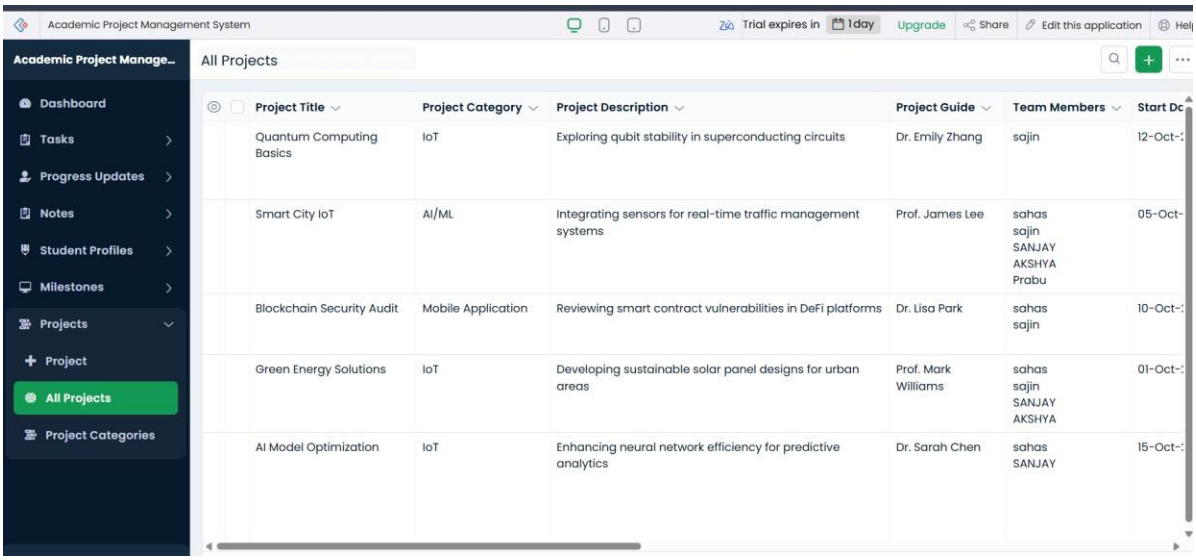


Figure 5.5: All Projects report showing project titles, categories, descriptions, guides and team members.

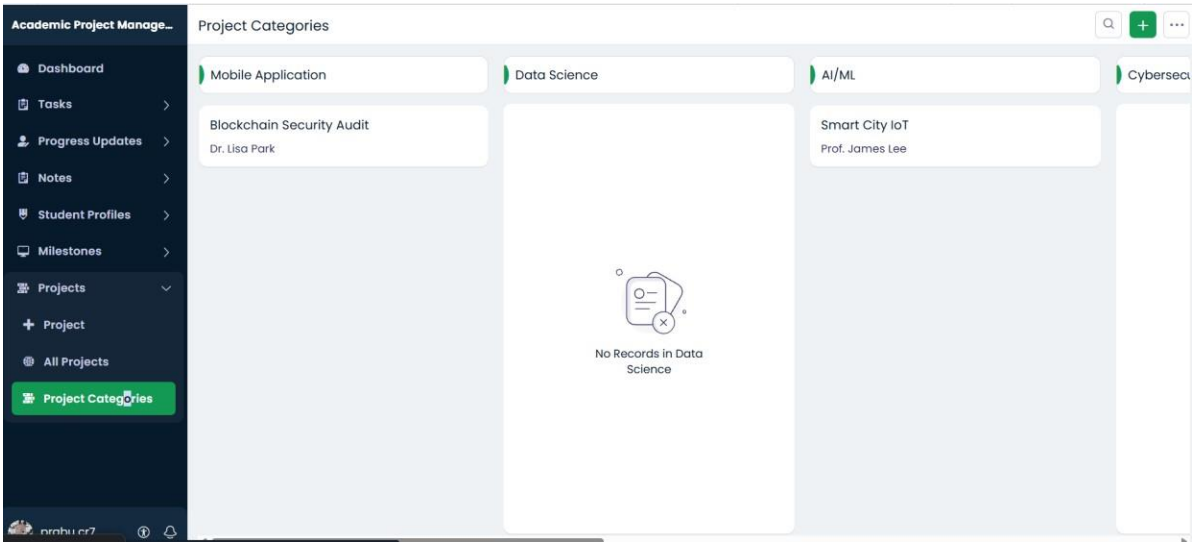


Figure 5.4: Project Categories board showing configured project categories.

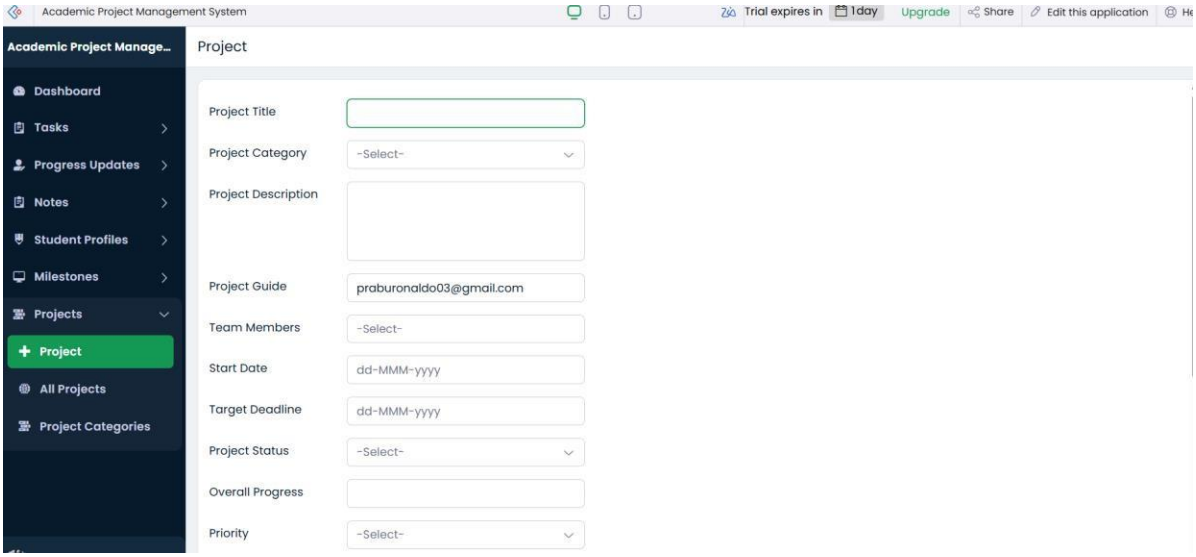


Figure 5.3: Project form for entering project details, guide, team, dates, status and priority.

Stores the project title, description, category, timeline, guide, team, technology stack and repository link where these fields are configured.

Task_Form

Stores task ownership, priority, due date, status, completion and blockers. Each task should be associated with its parent project.

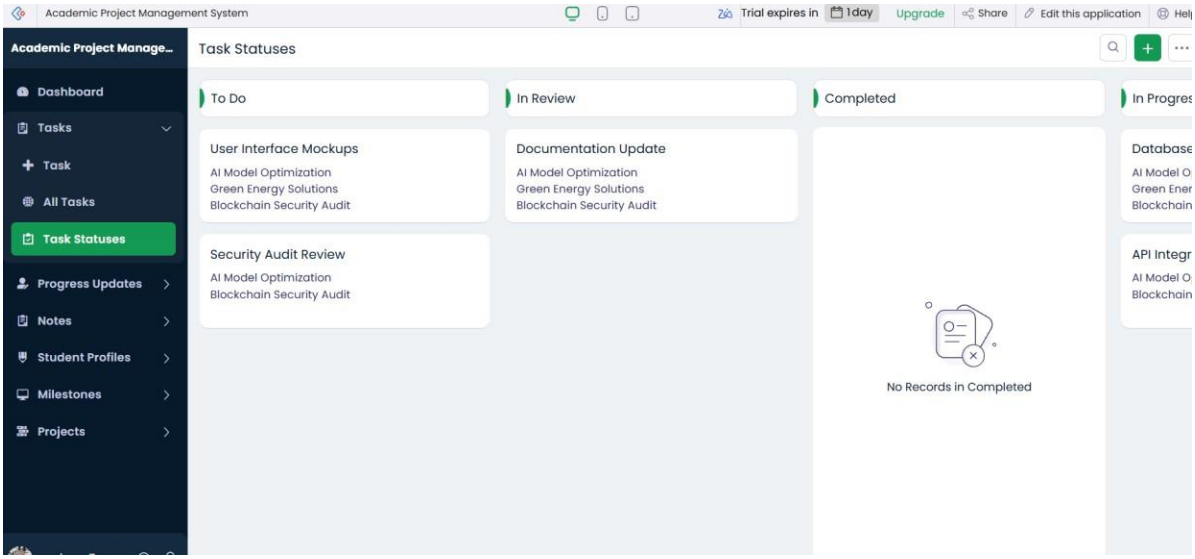


Figure 5.5: Task Statuses board grouped by status.

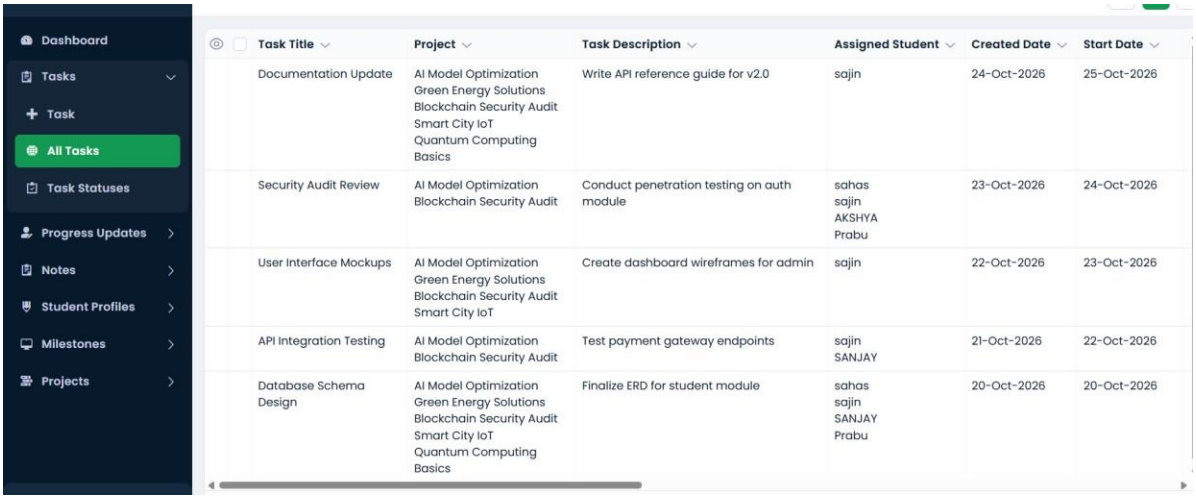


Figure 5.4: All Tasks report displaying saved task records.

Milestone_Form

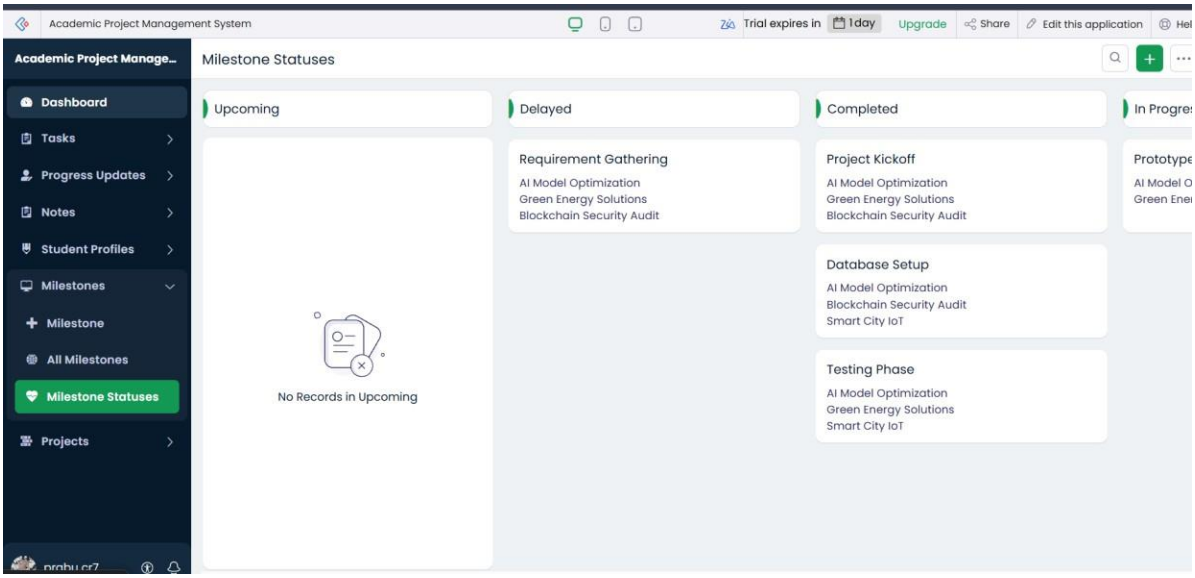


Figure 5.11: Milestone Statuses board grouping milestones by status.

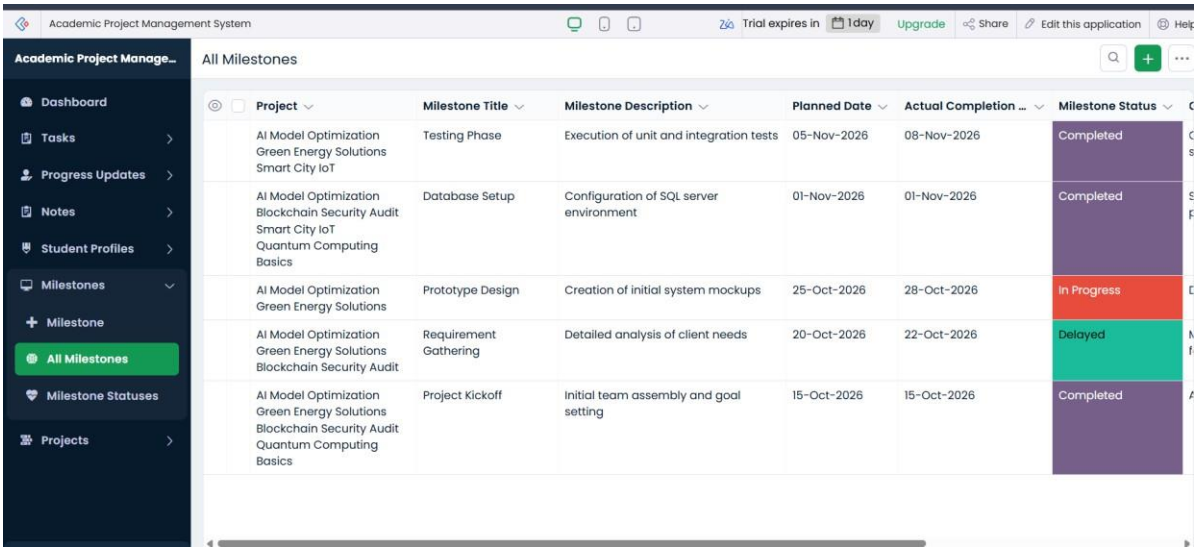


Figure 5.10: All Milestones report showing planned dates, completion dates and status.

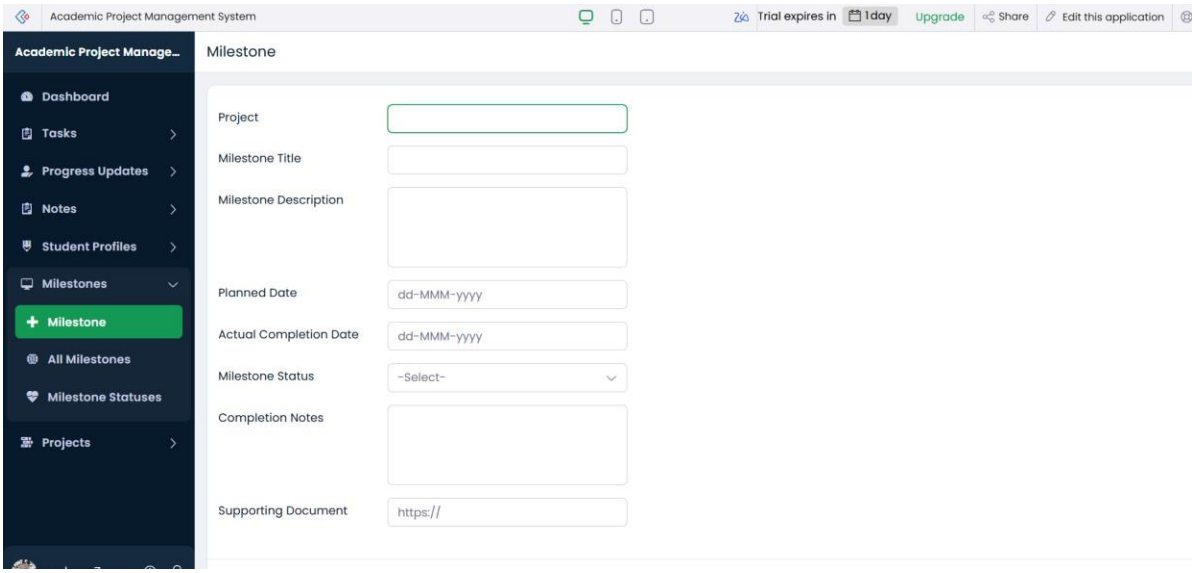


Figure 5.9: Milestone form for recording project milestone details and dates.

Captures significant project stages and their target dates and status.

Progress_Update_Form

Stores periodic updates about completed work, upcoming steps, challenges and assistance needed.

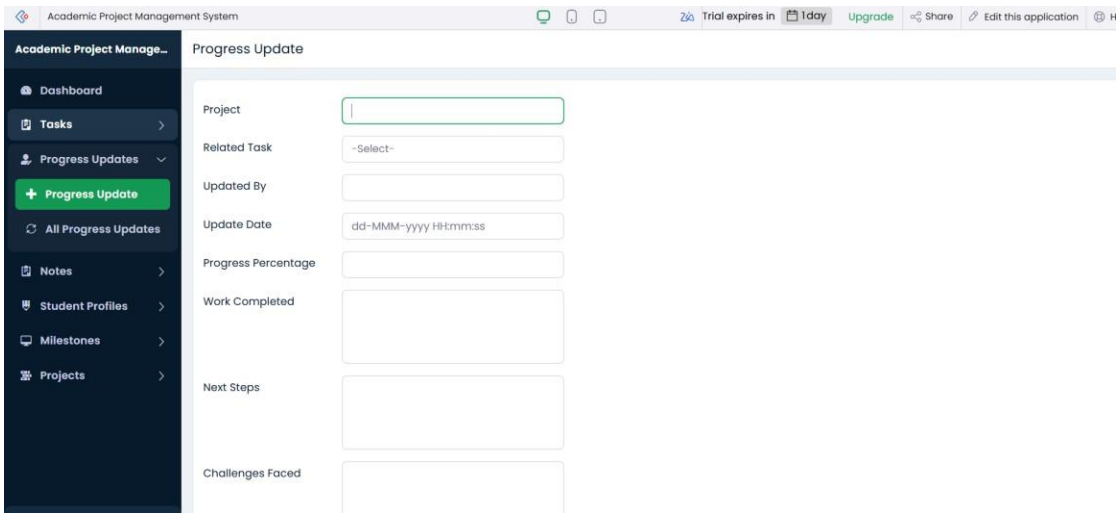


Figure 5.6: Progress Update entry form.

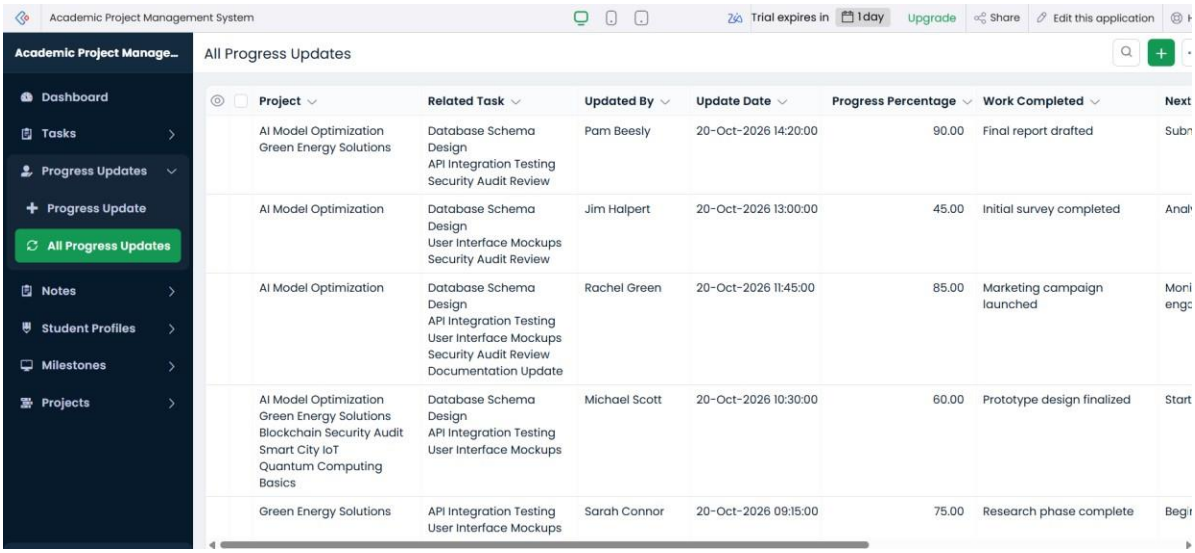


Figure 5.7: All Progress Updates report.

Meeting_Notes_Form

Stores meeting dates, attendees, decisions and action items for future reference.

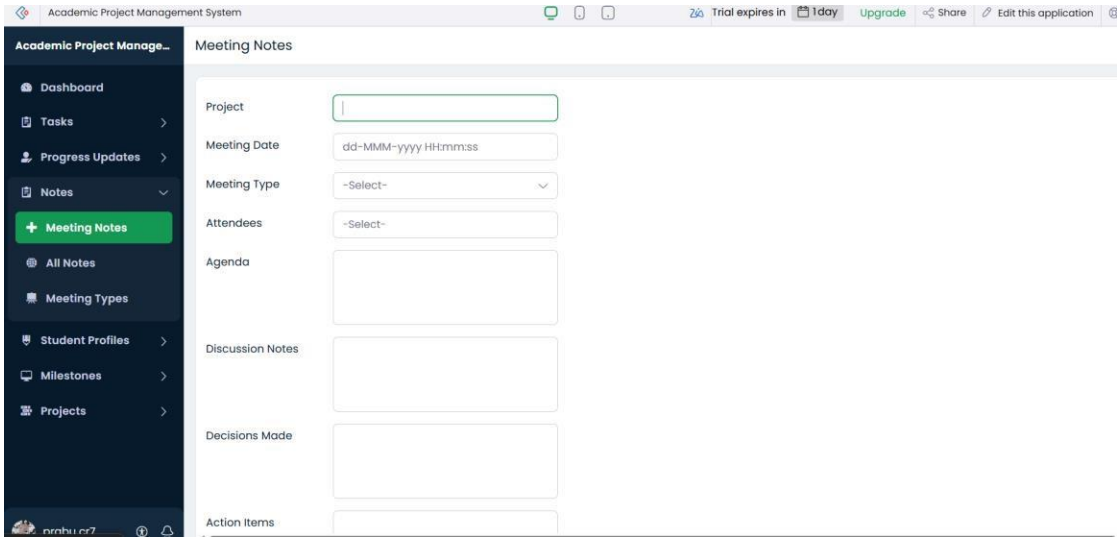


Figure 5.8: Meeting Notes entry form.

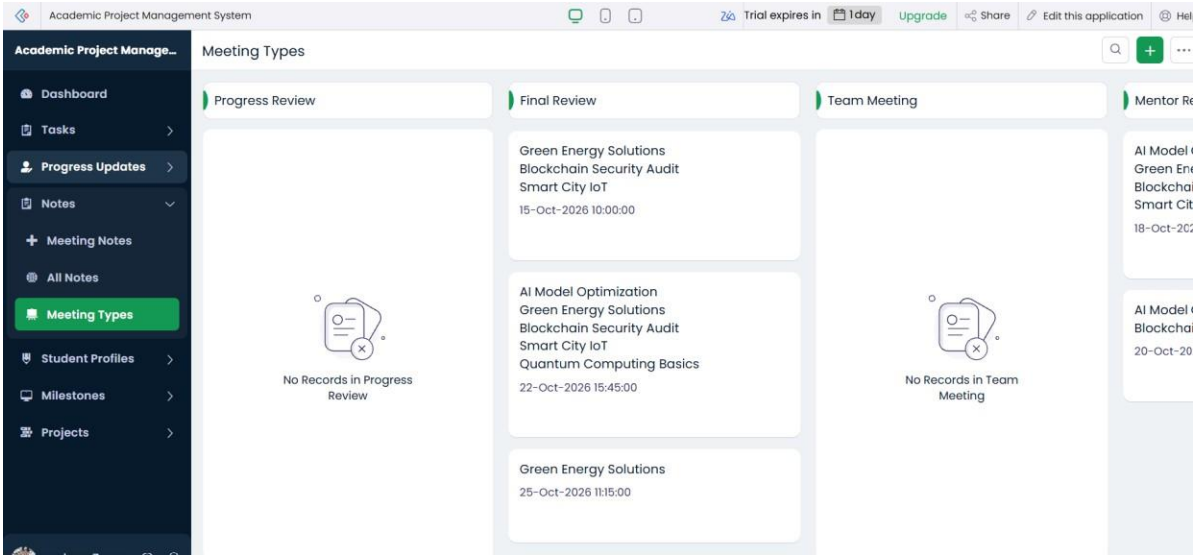


Figure 5.10: Meeting Types board.

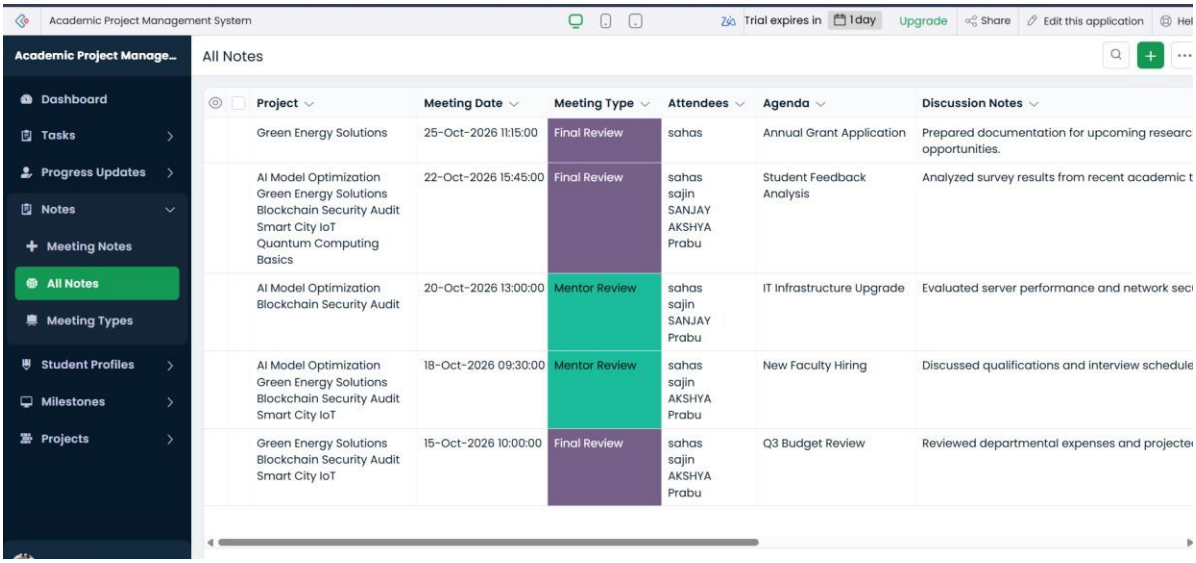


Figure 5.9: All Notes report with meeting records.

5.3 Reports

Reports should be created from the actual form records. Suggested reports include My Projects, My Tasks, Overdue Tasks, Upcoming Deadlines, Progress History and Project Milestones. Configure filters, sorting and grouping only after the required fields and relationships exist.

Report name	Type	Source form	Purpose / filter

My Projects	List	Project_Form	Projects associated with the logged-in student or team.
My Tasks	List / grouped	Task_Form	Tasks assigned to a selected student; group by status.
Overdue Tasks	Filtered list	Task_Form	Due date before today and status not Completed.
Upcoming Deadlines	Calendar or list	Task_Form / Project_Form	Tasks or projects due within a defined window.
Progress History	Chronological list	Progress_Update_Form	Sort update date in descending order.
Project Milestones	List / chart	Milestone_Form	Review upcoming, completed and delayed milestones.

5.4 Pages and Dashboards

Page	Main responsibility	Intended components
Student_Dashboard_Page	Overall summary	Project counts, pending tasks, approaching deadlines and recent activity.
Project_Management_Page	Manage project records	Project form and project report.
Task_Management_Page	Review and update tasks	Task report, filters and status grouping.
Project_Details_Page	View one project	Project details and related tasks, milestones and updates.
Progress_Tracking_Page	Record progress	Progress update form, history and milestone view.
Team_Workspace_Page	Coordinate the team	Team information, assignments, meeting notes and action items.
Reports_And_Analytics_Page	Analyse progress	Status charts, overdue tasks and deadline summaries.

Dashboard metrics must be calculated from actual records. For example, an overdue-task count should include tasks with a past due date whose status is not Completed. Any chart values shown in a demonstration should come from the application data, not illustrative numbers.

5.5 Deluge Scripting

Deluge scripts are optional extensions and should be documented only after they have been written and tested. The project description proposes date validation, progress calculation, overdue identification and deadline notifications. No exact script source was supplied in the project text, so this report does not invent executable Deluge code or claim that these scripts are operational.

Script ID	Proposed location	Purpose	Status to verify
DS-01	Project form validation	Check that target deadline is not earlier than start date.	Not verified
DS-02	Task form validation	Validate completion percentage and due-date consistency.	Not verified
DS-03	Workflow / function	Calculate overall project progress from task completion values.	Not verified
DS-04	Scheduled workflow	Identify overdue tasks and optionally notify responsible users.	Not verified

Insert the actual code and the real trigger location in Appendix C once the scripts exist. Do not include credentials, tokens or private connection details in source listings.

5.6 Workflows and Automation

Workflow	Type	Trigger / condition	Intended action	Status
Date validation	Form workflow	On validation / submit	Prevent inconsistent date values.	Verify in app
Progress calculation	Form workflow or function	Task progress changes	Recalculate project progress if configured.	Verify in app
Overdue identification	Scheduled workflow or report criteria	Due date has passed and task is incomplete	Flag or display overdue tasks.	Verify in app
Deadline notification	Scheduled / form workflow	Deadline is approaching	Notify assigned users if email settings are configured.	Verify in app

5.7 Integrations

The supplied project description does not confirm any external system integration. Zoho Creator forms, reports and internal workflows are part of the platform but should not be described as external integrations. If an integration is added later, document the service, purpose, connection method, authentication type and data exchanged. Never include secret keys or access tokens in the report.

Integration	Service	Method	Status
External API	None confirmed	Not applicable	Not evidenced in supplied project description.
Email notification	Optional Creator sendmail/workflow action	Only if configured	Test recipient, trigger and delivery before claiming.

5.8 User Management and Role-Based Access

The intended roles are Student, Project Guide and Administrator. Configure Creator users, profiles and record-level access criteria to match the role–permission matrix in Section 4.6. Test each role by signing in with an appropriate test account and attempting both permitted and restricted actions. Current configuration and access results should be captured in the screenshot placeholder below.

5.9 AI and Analytics Features

Zia AI is planned as an aid for creating or configuring supported application components. Using an AI-assisted builder does not by itself mean the finished application contains AI-powered project recommendations or predictive analytics. Initial analytics should rely on actual project, task and milestone records. AI-generated progress summaries or risk recommendations are possible future enhancements and require separate design and validation.

5.10 Pro-Code Extension

A separate pro-code extension is not confirmed in the supplied project description. The initial version is designed to remain within Zoho Creator, using configuration and Deluge where necessary. If future requirements exceed the platform's capabilities, a separate service or API integration could be evaluated; it is not part of the verified current scope.

5.11 Mobile and Portal Access

The current project description targets a web application and does not confirm a configured mobile-specific layout or student portal. Mobile and portal access should therefore be recorded as not verified unless it is enabled and tested. Any future mobile demonstration should show the actual application on the supported device or Creator mobile app.

CHAPTER 6

TESTING AND DEBUGGING

6.1 Testing Strategy

Testing should be performed after the forms, lookups, reports and workflows are configured. Form-level testing checks required fields and validation. Relationship testing checks that child records point to the correct project and student. Report testing checks filters and status calculations. Workflow testing checks trigger conditions and outcomes. Access testing checks the actual permissions for Student, Project Guide and Administrator accounts. Results in the table below should be updated from observed outcomes; do not mark a test as passed until it has been run.

6.2 Test Cases

TC ID	FR ref.	Test description	Steps / input	Expected result	Actual / status
TC-01	FR-01	Create a valid student profile	Enter all required fields and submit.	Record saves and appears in profile report.	To be tested
TC-02	FR-01	Submit profile with required field missing	Leave one mandatory field blank.	Validation message appears; invalid record is not saved.	To be tested
TC-03	FR-02	Create a project with valid dates	Enter start date and later deadline.	Project record saves.	To be tested
TC-04	FR-02	Use deadline before start date	Enter earlier deadline.	Validation prevents inconsistent dates if rule configured.	To be tested
TC-05	FR-03	Create task linked to project	Select project and assignee; submit.	Task is associated with selected project and student.	To be tested
TC-06	FR-03	Check task status report	Change a task status and open report.	Report reflects the saved status.	To be tested
TC-07	FR-04	Create milestone	Select project and enter milestone/date.	Milestone saves and appears under project.	To be tested
TC-08	FR-05	Submit progress update	Enter completed work and next steps.	Update saves and appears in progress history.	To be tested

TC-09	FR-06	Add meeting notes	Enter decisions and action items.	Meeting record is stored and linked to project.	To be tested
TC-10	FR-07	Open My Projects report	Use a test user with project records.	Only the expected project records appear.	To be tested
TC-11	FR-08	Check overdue task logic	Use past due date and non-completed status.	Task appears in overdue report if configured.	To be tested
TC-12	FR-08	Completed task with past due date	Set status to Completed.	Task is excluded from overdue list if criteria configured.	To be tested
TC-13	FR-09	Enter invalid completion percentage	Enter value outside allowed range.	Validation prevents or flags invalid value if configured.	To be tested
TC-14	FR-10	Student attempts restricted admin action	Sign in as Student and try to open admin-only data.	Access is denied by actual permissions.	To be tested
TC-15	FR-10	Guide reviews assigned project	Sign in as Guide and open assigned project.	Guide sees authorised project data only.	To be tested
TC-16	NFR-01	Review navigation clarity	Navigate through pages and forms.	Labels and navigation are understandable.	To be tested
TC-17	NFR-03	Review report performance	Open filtered report with representative data.	Report loads acceptably; record measured time if required.	To be tested
TC-18	Workflow	Test deadline notification	Use a test recipient and controlled trigger.	Notification is sent once under the configured condition.	To be tested

6.3 Debugging Deluge Scripts

When Deluge logic is implemented, debugging should use Creator's error messages, workflow execution history and supported logging statements such as info. Test one condition at a time with controlled records. Record the original symptom, root cause and correction. Since no actual script error or debug log was supplied, a specific defect is not claimed in this draft.

6.4 Defect Log

Bug ID	Description	Root cause	Correction	Status
BUG-01	[Enter actual defect found during testing]	[Determine from observed error]	[Record the fix and retest]	Open / Closed

BUG-02	[Optional additional defect]	[To be completed]	[To be completed]	Open / Closed

6.5 User Acceptance Testing

User acceptance testing should involve at least two representative users, ideally a student and a project guide. Ask each user to complete realistic tasks and record feedback. The entries below remain blank until feedback is collected.

User / participant	Role tested	Feedback	Action taken
[Participant 1]	Student	[Record actual feedback]	[Record action]
[Participant 2]	Project Guide	[Record actual feedback]	[Record action]

CHAPTER 7

DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application should be developed and tested before any production publication. If the Zoho Creator account supports separate Development, Stage and Production environments, changes should be reviewed and tested in the appropriate environment before release. The final report must identify which environments were actually used and the date of any production deployment. Publication or production deployment is not confirmed by the supplied project description.

S.No.	Deployment check	Status
1	Forms and lookup relationships reviewed	To be confirmed
2	Required-field and date validations tested	To be confirmed
3	Reports and dashboard metrics checked against real records	To be confirmed
4	Workflows tested with controlled test data	To be confirmed
5	User permissions checked for each role	To be confirmed
6	Test data reviewed and sensitive data removed where appropriate	To be confirmed
7	Production publication date and version documented	To be confirmed

7.2 Performance Optimisation

Performance should be reviewed using actual application data. Useful practices include filtering reports by relevant project or user, limiting unnecessary columns, avoiding repeated data retrieval inside loops in Deluge, and using appropriate lookup relationships. No measured before-and-after performance figures were supplied; therefore numerical improvements are not claimed.

Optimisation area	Potential action	Evidence to record
Reports	Apply meaningful filters and show only necessary columns.	Load time and report usability.
Deluge	Avoid repeated fetches inside loops where possible.	Execution behaviour and log results.
Data model	Use lookup fields for related entities.	Correct linked-record retrieval.

Dashboard	Use concise metrics and avoid duplicate components.	Accuracy of values and page responsiveness.

7.3 Security Features

Control	Application area	Purpose / verification
User authentication	Creator user access	Confirm that users must authenticate as required.
Role/profile permissions	Forms, reports and pages	Confirm each role can perform only authorised actions.
Record-level access	Projects and related records	Test that users cannot access unrelated project records.
Field-level access	Sensitive profile fields if any	Hide or restrict fields only through appropriate data permissions.
Credential handling	Connections / integrations	Do not store API keys or passwords in report text or client-visible fields.
Data minimisation	Student and project records	Collect only information needed for project tracking.

The existence of a permissions interface is not proof that access controls are correct. Security claims should be based on configuration review and negative tests.

7.4 Governance and Administration

Governance includes consistent naming of forms and fields, controlled changes to workflows, periodic review of user access, export or backup practices supported by the platform, and documentation of releases. The administrator should keep a record of configuration changes and ensure that demonstration data does not expose real student information. Any audit-trail or governance settings should be captured only if they are available and enabled in the application.

CHAPTER 8

RESULTS AND DISCUSSION

8.1 Application Output

The end-to-end demonstration should follow one realistic project scenario. A project is created with a guide and team, tasks and milestones are added, a student records progress, and the guide reviews the latest update and deadlines.

8.2 Achievement of Objectives

Objective	Implemented through	Outcome / evidence
Structured project records	Project and Student Profile forms	Confirm with actual form screenshots and saved records.
Task ownership and deadlines	Task form and task reports	Confirm task lookup, assignee, status and due-date behaviour.
Milestone and progress history	Milestone and Progress Update forms	Confirm saved records and project association.
Meeting records	Meeting Notes form	Confirm decisions and action items can be retrieved.
Project overview	Dashboard and reports	Confirm displayed values match actual records.
Validation and access control	Form rules and Creator permissions	Confirm with test results and settings screenshots.

8.3 Comparison with the Existing System

Parameter	Manual baseline	Proposed application	Evidence required
Information storage	Records may be spread across files and messages.	Related records are intended to be stored in one application.	Show forms and linked records.
Task visibility	Status requires manual follow-up.	Task reports can display status, owner and due date.	Show actual task report.
Progress history	Updates may remain in messages or meeting notes.	Progress updates can be stored chronologically.	Show saved progress records.
Deadline review	May rely on manual reminders.	Filtered reports or configured workflows can flag deadlines.	Show criteria and test results.
Reporting	May require manual compilation.	Reports and charts can summarise saved records.	Compare against underlying records.

8.4 Limitations of the Low-Code Approach

The application depends on the capabilities and constraints of Zoho Creator. Complex calculations or highly customised interfaces may require Deluge or a separate pro-code

extension. Analytics are only as accurate as the underlying records and configured criteria. Role-based security must be implemented through platform permissions, not merely by hiding interface components. Workflow actions and email notifications may require additional configuration and testing. These limitations should be reassessed against the final live application.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Form creation, field configuration, reports and page layout largely fit the no-code or visual low-code end of the spectrum. Lookup relationships, validation rules and workflow conditions require data-modelling and platform knowledge. Custom calculations and event-driven logic may use Deluge. A separate service, advanced analytics pipeline or custom client may require pro-code development if future requirements exceed Creator's built-in capabilities. The appropriate choice depends on complexity, maintainability, security and integration needs.

CHAPTER 9

CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Student Project Progress Tracker is designed to address common coordination challenges in academic project work by bringing project information, task assignments, milestones, progress updates and meeting decisions into a structured Zoho Creator application. The proposed six-form data model and seven-page navigation provide a foundation for recording project activity and reviewing progress. Reports can help students and project guides identify incomplete work, upcoming deadlines and project history without compiling information manually from separate sources.

The final effectiveness of the application depends on the completeness of the live configuration. Lookup relationships, dashboard metrics, workflow automation, access permissions and notifications must be tested before they are described as operational. The screenshot placeholders in this report are reserved for evidence from the finished application, and all implementation counts and deployment details should be updated to match the actual Zoho Creator configuration.

9.2 Future Enhancements

- Add a Kanban board for task status and a Gantt-style view for project timelines.
- Generate weekly progress summaries for students and project guides.
- Add mentor feedback, review rubrics and evaluation history.
- Explore risk indicators based on overdue tasks, blocked work and delayed milestones.
- Integrate repository activity or notification services if required and securely configured.
- Introduce AI-assisted progress summarisation after validating privacy, accuracy and explainability.
- Create reusable templates for AI/ML, IoT, web development and research projects.

9.3 Learning Outcomes and Career Relevance

This project provides practical experience in requirements analysis, data modelling, form design, lookup relationships, reports, dashboards, workflow design and application testing. It also highlights the importance of distinguishing an intended feature from a verified implementation. These skills are relevant to low-code development, business process automation, solutions consulting and application support roles. Further learning should focus on secure permissions, Deluge debugging, deployment governance and systematic testing.

REFERENCES

[1] Zoho Corporation, “Zoho Creator Help Documentation.” [Online]. Available: https://www.zoho.com/creator/help/. Accessed: 20 Oct. 2026.

[2] Zoho Corporation, “Deluge Script Reference.” [Online]. Available: https://www.zoho.com/deluge/. Accessed: 20 Oct. 2026.

[3] Zoho Corporation, “Zoho Creator Resources.” [Online]. Available: https://www.zoho.com/creator/resources/. Accessed: 20 Oct. 2026.

[4] W. Rosam and O. Kalnichevski, Low-Code Development: A Practical Guide. New Delhi, India: BPB Publications. Verify edition and publication year against the copy consulted before final submission.

[5] I. Sommerville, Software Engineering, 10th ed. Pearson, 2015. Use only if consulted for requirements and testing concepts.

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Mark the status only after checking the live Zoho Creator application.

S.No.	Zoho Creator Feature	Done?	Working in the Application?	Report Section	Figure / Screenshot Ref.
1	Application created and named	☐		5.1	
2	Student Profile form and required fields	☐		5.2	
3	Project form and project details	☐		5.2	
4	Task form, owner, priority and due date	☐		5.2	
5	Milestone form and target dates	☐		5.2	
6	Progress Update form	☐		5.2	
7	Meeting Notes form	☐		5.2	
8	Lookup relationships between records	☐		4.2	
9	Reports for projects and tasks	☐		5.3	
10	Dashboard / management pages	☐		5.4	
11	Field validation and mandatory fields	☐		5.5	
12	Deluge scripts, if implemented	☐		5.5	
13	Workflow rules and notifications, if implemented	☐		5.5	
14	Role permissions verified	☐		7.3	
15	Testing and live demonstration	☐		6.1	
16	Deployment / published URL verified	☐		7.1	

Total features implemented _____ / 16

APPENDIX B – COURSE OUTCOME MAPPING

The following mapping connects the project activities with the learning outcomes demonstrated through the Student Project Progress Tracker. Confirm the official course-outcome wording with the course handout before final submission.

Course Outcome	Outcome / Skill	Project Evidence	Report Reference
CO1	Analyse the problem and define functional and non-functional requirements	Problem statement, stakeholder analysis, requirement tables	Chapter 1, Chapter 3
CO2	Design forms, relationships, pages and reports using a low-code platform	Data model, lookup fields, page navigation and report design	Chapter 4, Chapter 5
CO3	Configure business logic and workflows where implemented	Validation rules, Deluge scripts and workflow configuration	Chapter 5, Appendix C
CO4	Test the application and evaluate its behaviour	Test cases, defect log and verification evidence	Chapter 6
CO5	Consider deployment, permissions, security and governance	Role-permission matrix and deployment checklist	Chapter 7
CO6	Demonstrate and document the completed application	Screenshots, feature checklist, project report and viva	Chapter 5, Appendix A, Appendix E–F

Table B.1: Project-to-Course Outcome Mapping

Note: The CO labels above are provisional placeholders, not official course outcome codes. Replace them with the exact outcomes supplied by the course instructor.

APPENDIX C – DELUGE SOURCE CODE

Insert the exact Deluge scripts from the live Zoho Creator application in the spaces below. Do not list a script as implemented unless it exists in the application and has been tested. If the project uses only built-in configuration, state that no custom Deluge script was implemented for that feature.

Reference	Script / Function	Source Code / Verification Note
C.1	Form validation / mandatory-field checks	[Paste verified script or write "Not implemented – built-in validation only."]
C.2	Task due-date / overdue logic	[Paste verified script or write "Not implemented."]
C.3	Progress calculation or status update	[Paste verified script or write "Not implemented."]
C.4	Workflow notification / reminder	[Paste verified script or write "Not implemented."]
C.5	Other implemented function	[Add function name and exact source, if applicable.]

Table C.1: Deluge Script Inventory

C.1 Form validation / mandatory-field checks

```
[Paste verified script or write "Not implemented – built-in validation only."]
```

C.2 Task due-date / overdue logic

```
[Paste verified script or write "Not implemented."]
```

C.3 Progress calculation or status update

```
[Paste verified script or write "Not implemented."]
```

C.4 Workflow notification / reminder

```
[Paste verified script or write "Not implemented."]
```

C.5 Other implemented function

```
[Add function name and exact source, if applicable.]
```

APPENDIX D – USER MANUAL

This user manual describes the intended usage flow. Menu names and controls should be updated to match the final application screenshots.

D.1 Accessing the Application

Open the Zoho Creator application using the verified application URL. Sign in with the account provided for the demonstration. Do not include passwords in this report. If the application is not published, access it through the authorised development environment.

D.2 Navigating the Home Page

Use the available navigation menu or dashboard tiles to open the relevant project, task, milestone, progress and meeting sections. Replace this description with the actual menu names shown in the final screenshot.

D.3 Creating a Student Profile

Open the Student Profile form, enter the required student details, review mandatory fields and submit the form. Confirm that the record appears in its associated report.

D.4 Creating a Project

Open the Project form and enter the project title, description and available planning details. Select related student or guide records only where the configured lookup fields are available. Save and verify the record.

D.5 Adding and Updating Tasks

Open the Task form, select the parent project and task owner where those fields exist, enter the task details and due date, and select the available status and priority. Save changes and confirm that the task report reflects them.

D.6 Recording Milestones and Progress

Create a milestone with its target date and current status. Use the Progress Update form to record completed work, next steps, blockers and support needed. Verify that each update is linked to the correct project.

D.7 Recording Meeting Notes

Enter the meeting date, participants, decisions and action items in the Meeting Notes form, using only fields present in the application. Review the saved record after submission.

D.8 Viewing Reports and Dashboard

Open the available reports or dashboard pages to review project records, task status, upcoming deadlines and progress history. Report filters and charts depend on the final configuration.

D.9 Troubleshooting

If a record cannot be submitted, check required fields and field formats. If a lookup does not show a record, verify that the related record exists and that the current user has permission to view it. Report incorrect calculations or missing notifications to the application administrator.

D.10 Security and Good Practice

Use only your authorised account, enter accurate data, avoid sharing login credentials, and sign out on shared devices. Test role permissions with authorised test accounts before presenting the application.

APPENDIX E – EVALUATION RUBRIC

Student Name: _____ Register No.: _____

Project Title: Student Project Progress Tracker Date: _____

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics (where used)	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit logs	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner: _____ Signature of External Examiner: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	[_____]	**Register No.**	[_____]
Department / Section	AI & ML / [_____]	**Date of Viva**	[_____]
Application URL Verified	Yes / No	**Demo Duration**	_____min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1	Explain the problem solved by the Student Project Progress Tracker.		E / G / S / NI	
2	Why was Zoho Creator selected for this project?		E / G / S / NI	
3	Explain the purpose of the six planned forms and their relationships.		E / G / S / NI	
4	What is a lookup field and how does it connect related records?		E / G / S / NI	
5	How can the application identify overdue tasks?		E / G / S / NI	
6	What is Deluge, and which implemented feature uses it?		E / G / S / NI	
7	How would you verify permissions for students, guides and administrators?		E / G / S / NI	
8	Which tests should pass before the application is demonstrated or deployed?		E / G / S / NI	

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Name: _____ Name: _____